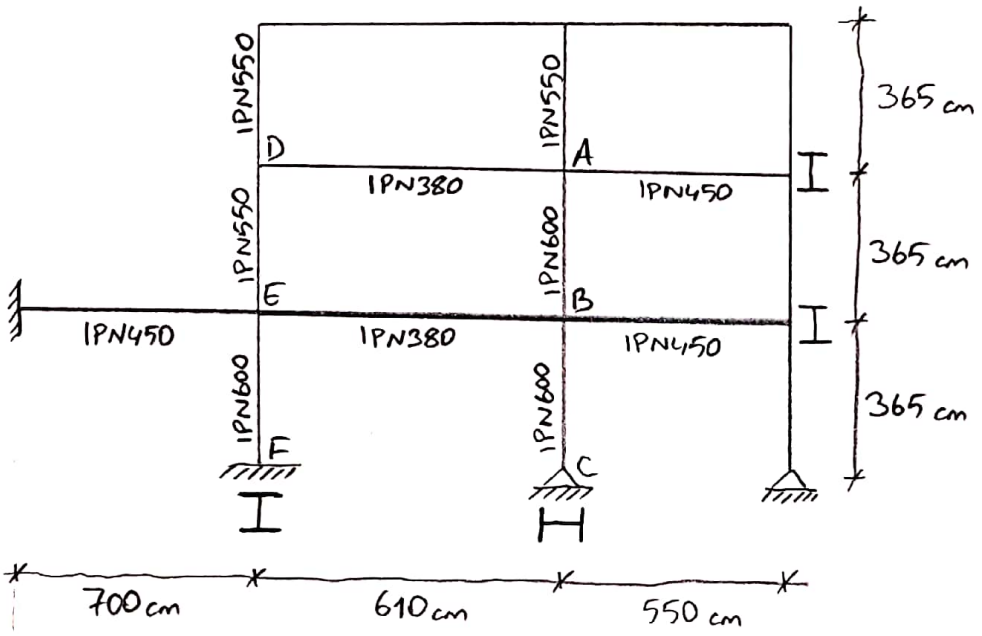


Determine effective length factors (K) of AB, BC, DE and EF columns.

AB, BC, DE ve EF kolonlarının burkulma boyu katsayılarını (K) hesaplayınız.

IPN 380	IPN450	IPN550	IPN600
$I_x = 24010\ cm^4$	$I_x = 45850\ cm^4$	$I_x = 99180\ cm^4$	$I_x = 138800\ cm^4$
$I_y = 975\ cm^4$	$I_y = 1730\ cm^4$	$I_y = 3490\ cm^4$	$I_y = 4674\ cm^4$
$A = 107\ cm^2$	$A = 147\ cm^2$	$A = 212\ cm^2$	$A = 254\ cm^2$

AB alımı (Sidesway allowed) (Yanal hareket önlenmemiş)

$$G_A = \frac{99180/365 + 138800/365}{24010/610 + 45850/550} = 5,313$$

$$G_B = \frac{138800/365 + 138800/365}{24010/610 + 45850/550} = 6,197$$

$$K = \sqrt{\frac{5,313.(1,6.6,197+4)+(4.6,197+7,5)}{5,313+6,197+7,5}} = 2,36$$

CamScanner ile tarandı

BC column (Sidesway prevented) (Yanal hareket önlenmiş)

$G_B = 6,197$

$G_C = 10$ (simply supported) (6.4.3 (a))
(basit mesnetli)

$$K = \frac{3.6,197.10 + 1,4.(6,197+10) + 0,64}{3.6,197.10 + 2.(6,197+10) + 1,28} = 0,953$$

DE column (Sidesway allowed) (Yanal hareket önlenmemiş)

$$G_D = \frac{3490/365 + 3490/365}{24010/610} = 0,486$$

$$G_E = \frac{3490/365 + 4674/365}{\frac{2}{3}.(45850/700) + 24010/610} = 0,269$$

(6.4.3.(c))

$$K = \sqrt{\frac{0,486.(1,6.0,269+4) + (4.0,269+7,5)}{0,486+0,269+7,5}} = 1,14$$

EF column (Sidesway prevented) (Yanal hareket önlenmiş)

$$G_E = \frac{3490/365 + 4674/365}{2.(45850/700) + 24010/610} = 0,131$$

(6.4.3.(b))

$G_F = 1$ (fixed support)
(ankastre mesnet)

$$K = \frac{3.0,131.1 + 1,4.(0,131+1) + 0,64}{3.0,131.1 + 2.(0,131+1) + 1,28} = 0,665$$

CamScanner ile tarandı